

Practical 0

Aim:- Introduction to DBMS Lab.

Theory:- A Database Management System (DBMS) is a software system that enables users to efficiently store, organize, manage, and retrieve data in a structured manner. It acts as an interface between the user and the database, ensuring that data is handled securely, accurately and efficiently. DBMS reduces data redundancy, maintains data integrity, and allows multiple users to access data simultaneously without conflicts.

In a DBMS, data is stored in the form of tables (relations) consisting of rows (tuples) and columns (attributes). Each table represents an entity, and relationships between tables are established using keys such as Primary Key, and Foreign Keys. This structured approach helps in maintaining consistency and avoiding duplication of data.

The DBMS lab focuses on practical implementation using SQL (Structured Query Language), which is the standard language used to interact with databases. SQL commands are broadly classified into the following categories:-

① DDL (Data Definition Language):- Used to define and modify the structure of database.

objects.

- ② DML (Data Manipulation Language) :- Used to insert, update, and delete data in tables.
- ③ DQL (Data Query Language) :- Used to retrieve data from the databases.
- ④ DCL (Data Control Language) :- Used to control access permissions.
- ⑤ TCL (Transaction Control Language) :- Used to manage transactions in the databases.

The lab also emphasizes the use of constraints to ensure data accuracy and reliability.

- ① Primary Key :- Uniquely identifies each record in a table.
- ② Foreign Key :- Establishes relationship between tables.
- ③ Not Null :- Ensures a column cannot have empty values.
- ④ Unique :- Prevents duplicate values.
- ⑤ Check :- Ensures values satisfy a specific condition.

The DBMS lab helps in understanding concepts like:-

- * Table creation and data handling.
- * Relationships between table (ER Model)
- * Normalization to reduce redundancy.
- * Writing efficient SQL queries.

* Conclusion:- The DBMS lab provided knowledge of how databases are created and managed using SQL. It helped in understanding how data is stored, retrieved and manipulated efficiently.

Through various practicals, we are going to learn to apply concepts like table creation, constraints, normalization and query execution. The lab also improved our problem-solving skills and understanding of real-world databases applications.

Practical 1

* Aim:- Introduction SQL and Oracle Installation.

* Theory:- SQL (Structured Query Language) is a standard programming language used to manage and manipulate relational databases. It allows users to perform various operations such as creating databases structures, inserting data, updating records, deleting data, and retrieving information efficiently. SQL is widely used because of its simplicity, flexibility, and powerful querying capabilities.

SQL commands are categorized into different types :-

- ① DDL (Data Definition Language) :- Used to define databases structure (CREATE, ALTER, DROP)
 - ② DML (Data Manipulation Language) :- Used to modify data (INSERT, UPDATE, DELETE)
 - ③ DQL (Data Query Language) :- Used to retrieve data (SELECT).
 - ④ DCL (Data Control Language) :- Used for access control (GRANT, REVOKE).
 - ⑤ TCL (Transaction Control Language) :- Used to manage transactions (Commit, ROLLBACK)
- SQL works with relational databases

where data is stored in tables consisting of rows and columns. It supports constraints such as Primary Key, Foreign Key, Unique and Not Null to ensure data integrity.

Oracle databases is a popular Relational Database Management System (RDBMS) developed by Oracle Corporation. It is widely used in industries for handling large-scale data and supports SQL for database operations.

* Oracle Database installation is the process of setting up the Oracle environment on a system so that databases can be created and managed. The basic steps involved are:-

① Download Oracle Software:-

Download the required version of Oracle Database from the official Oracle website.

② Run Installer :-

Execute the setup file ~~and~~ and launch the Oracle Universal Installer.

③ Choose Installation Type:-

Select option such as "Create and configure a Database" or "Install Software Only".

④ System Configuration:-

Set administrative passwords, choose database name (SID), and configure storage.

⑤ Install and Configure

The installer copies files, sets up the database, and configures necessary services.

⑥ Verification:-

After installation, tools like SQL*Plus or Oracle SQL Developer can be used to verify and interact with the database.

Oracle provides features like high security, scalability, backup and recovery, and multi-user access, making it suitable for enterprise-level applications.

* Conclusion:- The study of SQL and Oracle installation provides a strong foundation for understanding database systems. SQL enables efficient communication with databases, while Oracle offers a powerful platform to implement and manage them.

Through this practical, we learned how SQL is used for performing various database operations and how Oracle Database is installed and configured. This knowledge is essential for working with real-world applications where data management plays a crucial role.

Practical 2

* Aim:- Perform the following:-

- (a) Creating a Database
- (b) Viewing all databases
- (c) Creating Tables (With and Without Constraints).
- (d) Viewing all Tables in a Database
- (e) Inserting / Updating / Deleting Records in a Table,
- (f) Save point, Saving (Commit) and Undoing (rollback).

* Theory:- A Database Management System (DBMS) is used to store, organize and manage data in a structured form. SQL is the standard language used to perform operations on databases such as creating databases, tables and manipulating data.

- ① Create database rishabh;
Show databases;
use rishabh;

First, a database rishabh was created using Create database and selected using the USE statement. The available databases were displayed using Show Databases

- ② Create Table Student(

```
SID INT,  
Sname VARCHAR (20),  
Branch VARCHAR (20),  
city VARCHAR (20)  
);
```

A table student was created without constraints using the Create Table command. The varchar variables are created inside it.

```
③ INSERT INTO student VALUE (1, 'Rishabh Dwivedi',  
    'MCA', 'Nagpur');  
INSERT INTO student VALUE (2, 'Aman Singh',  
    'MCA', 'Nagpur');  
SELECT * FROM student;
```

Data was inserted into the table using INSERT statement, and records were displayed using the Select command.

```
④ Update student  
    SET city = 'Pune'  
    WHERE SID = 2;  
Delete FROM student  
    WHERE SID = 3;
```

Update command was used to modify existing records.

Delete command was used to remove specific records from the table.

⑤ CREATE TABLE Student Constrained (
SID INT PRIMARY Key;
Sname VARCHAR(20) NOT NULL,
Branch VARCHAR(20),
city VARCHAR(20));

Another table was created with constraints such as:-

- * Primary Key on SID (ensures uniqueness)
- * Not Null on Sname (prevents empty values)

⑥ START Transaction

```
INSERT INTO Student Values (4, 'Pranit Bhairam',  
'MCA', 'Balaghat');  
SAVEPOINT sp1;  
INSERT INTO Student Value (5, 'Shubham', 'BCCA',  
'Nagpur');  
Rollback to sp1;  
COMMIT
```

- * Start transaction was used to begin a transaction
- * Savepoint was used to mark a point within the transaction
- * Additional record were inserted after the savepoint
- * Rollback to savepoint was used to undo changes made after the savepoint
- * This demonstrated how partial changes can

be reversed without affecting earlier data

- * Commit command is used to save all the changes made in the current transaction permanently in the databases.

- * Conclusion:- In this practical, we learned how to create and manage databases and tables using SQL. We performed operations like inserting, updating and deleting records. We also understood transaction control using savepoints, commit, and rollback, which helps in maintaining data accuracy and consistency.

Practical 3

- * Aim:- Perform the following :- Data constraints (Primary key, Foreign key, Not Null) Data insertion into a table.
- * Theory:- A database management system (DBMS) uses constraints to maintain data accuracy, consistency, and integrity. Constraints are rules applied on table columns to ensure valid data entry.
 - ① Primary key:- A primary key uniquely identifies each record in a table.
 - * It does not allow duplicate values.
 - * It does not allow NULL values.
 - Ex:-
DeptID INT PRIMARY KEY
SID INT PRIMARY KEY
 - ② Not Null Constraints:- The NOT NULL constraint ensures that a column cannot have empty or NULL values.
Ex:-
DeptName VARCHAR(50) NOT NULL
Sname VARCHAR(20) NOT NULL
This ensures that every record must have a valid name.
 - ③ Foreign key:- A foreign key is used to create a relationship between two tables. It ensures that values in one table must exist in another table.

Ex:- FOREIGN KEY (DeptID) REFERENCES
Department (DeptID)

This creates a link between students & their departments.

Insert command was used to insert values in tables Department & Student-Info.

Data from both tables was displayed using:-
SELECT * FROM Department;
SELECT * FROM Student-Info;

* Conclusion:- In this practical, we learned how to apply Primary key, Foreign key and Not Null constraints to maintain data integrity. We also inserted and retrieved data while ensuring proper relationships between tables. This helped us understand the importance of constraints in database management.